

Introduction

The proposal is **audacious**.

It suggests that the United States should consider creating a national information architecture capable of maintaining a trustworthy longitudinal history for each participating patient—drawing information from healthcare organizations that today operate independently across a fragmented healthcare system.

That history could encompass decades of diagnoses, clinical notes, laboratory results, imaging, medications, procedures, treatments, claims, authorizations, and outcomes. Information would have to be matched to the correct patient, reconciled when sources disagree, preserved with its provenance, protected from unauthorized use, and made available in the context required for a particular decision.

Artificial intelligence would play two important roles.

1. After a patient encounter or new result, AI could help extract clinical meaning, reconcile it with the existing history, preserve unresolved questions, capture outcomes, and integrate the new information into a trustworthy longitudinal patient history while preserving links to the underlying evidence
2. When a clinical event occurs, AI could retrieve and summarize the relevant knowledge—including unresolved questions, risks, gaps, and prior outcomes—and relate it to treatments and outcomes for similar patients. It could also incorporate relevant clinical guidelines, research, and other external medical knowledge.

The federal government is moving in this direction. TEFCA, the CMS Health Technology Ecosystem, and CDC data-modernization initiatives are expanding interoperability, patient access, and the ability to transform healthcare and public-health data into usable information. These efforts are making healthcare information more accessible, **but access is not the same as integration.** They do not yet provide the continuously maintained, evidence-linked longitudinal patient history proposed in this book.

The objective is not to replace the judgment of physicians, payers, or other healthcare professionals. It is to give them better information on which to exercise that judgment.

- A physician should be able to see what has already been tried, what is known about the outcome, what remains unresolved, what risks or changes deserve attention, and what approaches have produced better outcomes for similar patients.
- A payer considering an authorization should have the clinical context needed to understand why a treatment is being requested, along with evidence on the costs and outcomes of alternative approaches.

The goal is better-informed decisions: the right care at the right time, with less delay, duplication, and administrative friction.

This is not only a clinical information problem. Healthcare is also commerce. Patients choose providers and treatments. Providers recommend and deliver care. Payers authorize and pay for it. Yet these decisions are often made without timely information about availability, experience, cost, alternatives, and outcomes.

The potential economic opportunity is enormous. U.S. healthcare spending is measured in trillions of dollars. Administrative costs, improper payments, repeated work, delayed decisions, and care delivered without relevant history create opportunities measured in billions. **Even relatively small improvements could produce substantial economic value.**

But the scale of the opportunity is not evidence that this proposal is feasible.

Creating such an infrastructure raises substantial technical, economic, privacy, legal, and political questions. This book does not claim to have resolved them or recommend beginning with a massive government technology program.

It recommends something much smaller: **determine whether the idea is worth pursuing.**

The following proposal describes a six-person, 60-day feasibility assessment to determine whether the idea warrants further investigation.

From Vision to Action: The Proposal

A 60-Day Feasibility Assessment

The proposal is deliberately ambitious: a national healthcare information architecture that could reduce costs, improve decisions, and make healthcare information more useful. The scale of the opportunity is substantial, but so are the technical, economic, legal, and political questions. The appropriate next step is not implementation. It is a focused, 60-day assessment to determine whether the potential benefits justify further action.



FEASIBILITY ASSESSMENT

DURATION: 60 DAYS

TEAM: 6 EXPERTS

Prescription:



DETERMINE:

1 Is it technically feasible?

2 Is the potential value significant?

3 What are the major barriers?



Prescribed by:

The Future of Better Healthcare

___/___/___

Date:

___/___/___



NEXT STEP: EVIDENCE BEFORE IMPLEMENTATION

From Vision to Action: The Proposal

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Purpose of the Assessment

The proposal described in this book is deliberately ambitious. It suggests that a national longitudinal healthcare information architecture could make existing healthcare information more useful—reducing administrative effort, improving decisions, and creating new opportunities to understand the cost and effectiveness of healthcare.

The next step should not be implementation. Nor should it begin with a large committee, procurement effort, or technology demonstration.

It should begin by answering three questions:

1. **Could better healthcare information create substantial measurable value?**
2. **Is an architecture capable of delivering that information technically feasible?**
3. **What are the most significant barriers to making it work?**

A focused 60-day assessment by six people with complementary expertise could provide enough evidence to determine whether the concept warrants further investigation.

The six-person team should combine expertise in healthcare delivery and clinical workflow; payer operations, claims, and payment; healthcare information systems and interoperability; data architecture and AI; privacy, security, and healthcare law; and healthcare economics and public policy. The team need not represent every healthcare constituency. Its role is to conduct the assessment, gather evidence from a much broader range of participants, and determine whether the opportunities identified are valuable, technically feasible, and reasonably addressable. Where specialized knowledge is required, the team should obtain it through the interviews rather than expanding the core assessment team.

Start With the Opportunity, Not the Architecture

The preceding chapters present one possible architecture. The feasibility assessment should not begin by asking healthcare experts whether they agree with it.

Doing so would constrain the discussion to the proposed solution.

Instead, the assessment should first discover what knowledgeable people throughout healthcare would do **if information limitations that constrain them today could be overcome.**

Healthcare professionals routinely work around fragmented, incomplete, delayed, difficult-to-interpret, or unavailable information. Some of those limitations create visible problems. Others consume substantial time and effort and have become accepted as part of delivering or paying for healthcare. Still others prevent potentially valuable things from being done at all.

The assessment should identify those opportunities before determining whether the proposed architecture—or some variation of it—could address them.

Prepare Interviewees to Think Beyond Today's Constraints

Interviewees should receive a short description of the assessment several days before the discussion. It should explain the objective without describing the proposed architecture in enough detail to influence their answers.

They might be told:

We are exploring whether better use of healthcare information could materially improve healthcare decisions, operations, costs, and outcomes.

Before the discussion, think about your own area of healthcare—not about a particular technology or proposed solution. Consider the important problems you encounter, opportunities that cannot be pursued today, and activities that consume substantial time or effort because the information needed for a decision is difficult to obtain, interpret, or trust.

Then imagine those information limitations did not exist. If the information or knowledge you wished for were available, trustworthy, and timely, what could you do differently?

Don't limit your thinking to what today's systems can provide.

The objective is to get interviewees to step outside current processes and systems long enough to consider what might be possible.

Structured Discovery Interviews

The assessment team should conduct focused interviews with knowledgeable representatives of healthcare providers, payers, healthcare information technology, interoperability, government, employers, researchers, and other appropriate participants.

The interviews need not be long. Thirty to sixty minutes should be sufficient if participants have considered the questions in advance.

Four questions should guide the discussion:

1. **What are the three to five most important problems, missed opportunities, or areas of significant effort in your part of healthcare?**

For each item described, ask:

2. **What information, knowledge, or capability do you wish you had that would help address it?**
3. **If that were available, what could you do differently?**
4. **What measurable value could result?**

The interviewer should probe for specific examples without steering the participant toward the proposed architecture. If the participant cannot estimate the overall value, ask for a specific example, the value associated with that instance, and how often similar situations occur. These examples can provide both a rough estimate of potential value and concrete evidence for the final assessment.

The objective is to establish a chain from:

Problem / Opportunity / Effort → Needed Information or Capability → Different Action → Measurable Value

Different participants will define value differently. It might include reduced administrative effort, faster decisions, fewer repeated tests, better treatment choices, reduced payment errors, shorter authorization times, better outcomes, improved patient experience, or opportunities not anticipated in this book.

Consolidate What Healthcare Says It Needs

The interviews should then be consolidated across organizations and disciplines.

AI could help analyze interview transcripts, identify recurring problems and desired capabilities, group similar opportunities, and find relationships that might otherwise be missed. The assessment team—not AI—would determine the meaning and significance of the findings.

For each significant opportunity, the team should document:

- the problem, missed opportunity, or effort being addressed;
- the information or capability participants said they need;
- what could be done differently if it were available;
- who would benefit;
- how the benefit could be measured; and
- the credible range of potential value.

This creates an evidence-based description of what healthcare participants say they could do with better information, independent of the architecture proposed in this book.

Then Compare the Needs With What Already Exists

Only after identifying the opportunities should the assessment turn to architecture.

The team should determine which healthcare information systems, TEFCA and QHINs, FHIR, IHE profiles, other interoperability initiatives, commercial products, or existing government and industry infrastructure already provide required capabilities.

The objective should not be to justify building something new. **If an existing capability already solves part of the problem, use it.**

The analysis should identify:

Needed Capability → What Exists Today → What Is Missing → What, If Anything, Must Be Added

This step may confirm parts of the architecture proposed in this book, modify them, or demonstrate that some proposed capabilities are unnecessary.

Assess Technical Feasibility

The remaining gaps can then be evaluated technically.

Could participating systems reliably communicate relevant new, changed, and deleted information? Can information describing the same patient and clinical events be matched across organizations? Can AI extract clinical meaning while preserving the underlying evidence? Can duplicate and conflicting information be identified and reconciled without concealing uncertainty? Can provenance and temporal history be preserved? Can relevant longitudinal knowledge be retrieved quickly enough to support healthcare decisions?

The assessment does not need to design the system. It needs to determine whether credible approaches exist for the major technical requirements and identify requirements for which feasibility remains uncertain.

Determine Whether the Potential Value Is Significant

Technical feasibility alone is insufficient.

The assessment should estimate the potential value of the opportunities identified through the interviews. Precision is neither possible nor necessary in 60 days. The objective is to determine whether credible benefits appear large enough to justify additional work.

Some benefits may be directly economic: administrative labor, duplicate services, payment errors, prior-authorization effort, or avoidable utilization.

Others may require different measures: physician preparation time, decision latency, patient burden, treatment effectiveness, safety, continuity of care, or improved ability to compare treatments and providers.

The important question is not whether every proposed benefit can be quantified. It is whether the combined evidence suggests an opportunity substantial enough to warrant further investigation.

Challenge the Findings

Before reaching a conclusion, the team should make the strongest case **against** proceeding.

The assessment should examine privacy, security, patient consent, governance, legal authority, patient identification, data quality, operational reliability, adoption, incentives, cost, concentration of risk, and other significant concerns identified during the work.

Classify each concern as a fundamental barrier, an unresolved question requiring further work, or a risk that could reasonably be addressed through architecture, governance, legislation, auditing, or implementation practices.

The purpose is not to defend the proposal. It is to determine whether it survives serious scrutiny.

The 60-Day Decision

At the end of 60 days, the assessment should answer the questions with which it began:

Is the potential value significant?

Is the required architecture technically feasible?

Are the major barriers reasonably addressable?

The result does not need to be simply yes or no.

It may conclude

1. **Stop** — The potential value does not justify further work.
2. **Resolve** — The opportunities appear promising, but specific questions must be resolved before determining whether to proceed.
3. **Proceed** — The potential benefits and technical feasibility justify a more detailed architectural and economic assessment.

In none of those cases does the 60-day assessment commit the government to a pilot, procurement, implementation program, or particular architecture.

Its purpose is much simpler: to determine whether the opportunity is important enough—and credible enough—to take the next step.

A Decision, not a Commitment

Healthcare has no shortage of committees studying broad problems.

This proposal calls for something smaller and more disciplined: six qualified people, 60 days, and a specific question.

Can a secure national longitudinal health information architecture give providers timely access to relevant patient history, reduce administrative friction, and create measurable benefits for patients, providers, and payers?

If the answer is no, document the reasons.

If the answer is yes, the nation can then decide what—if anything—to do next.